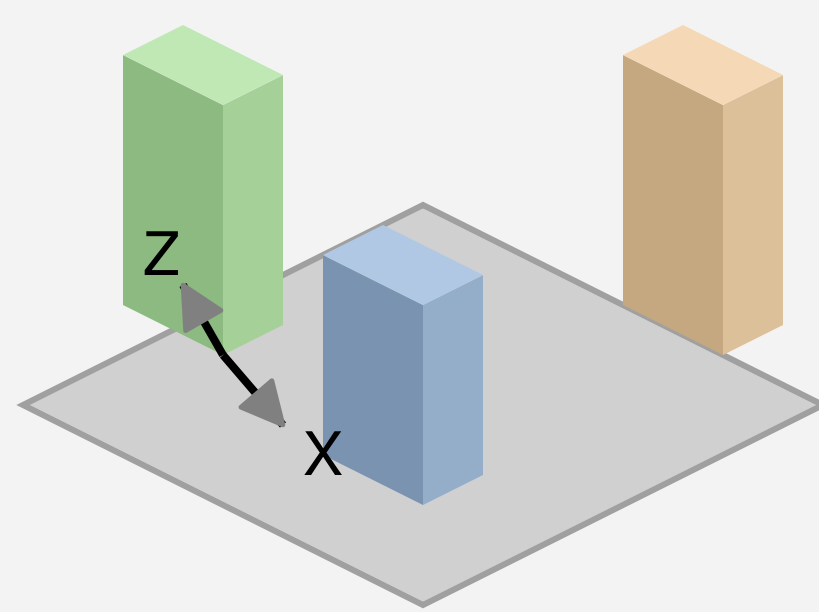


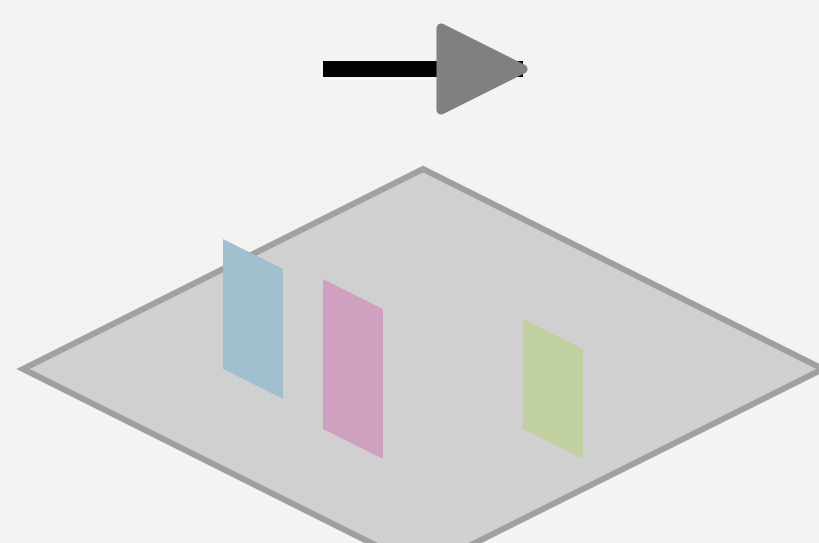
3. Hierarchical Planning for VLM-Guided Assembly

Input

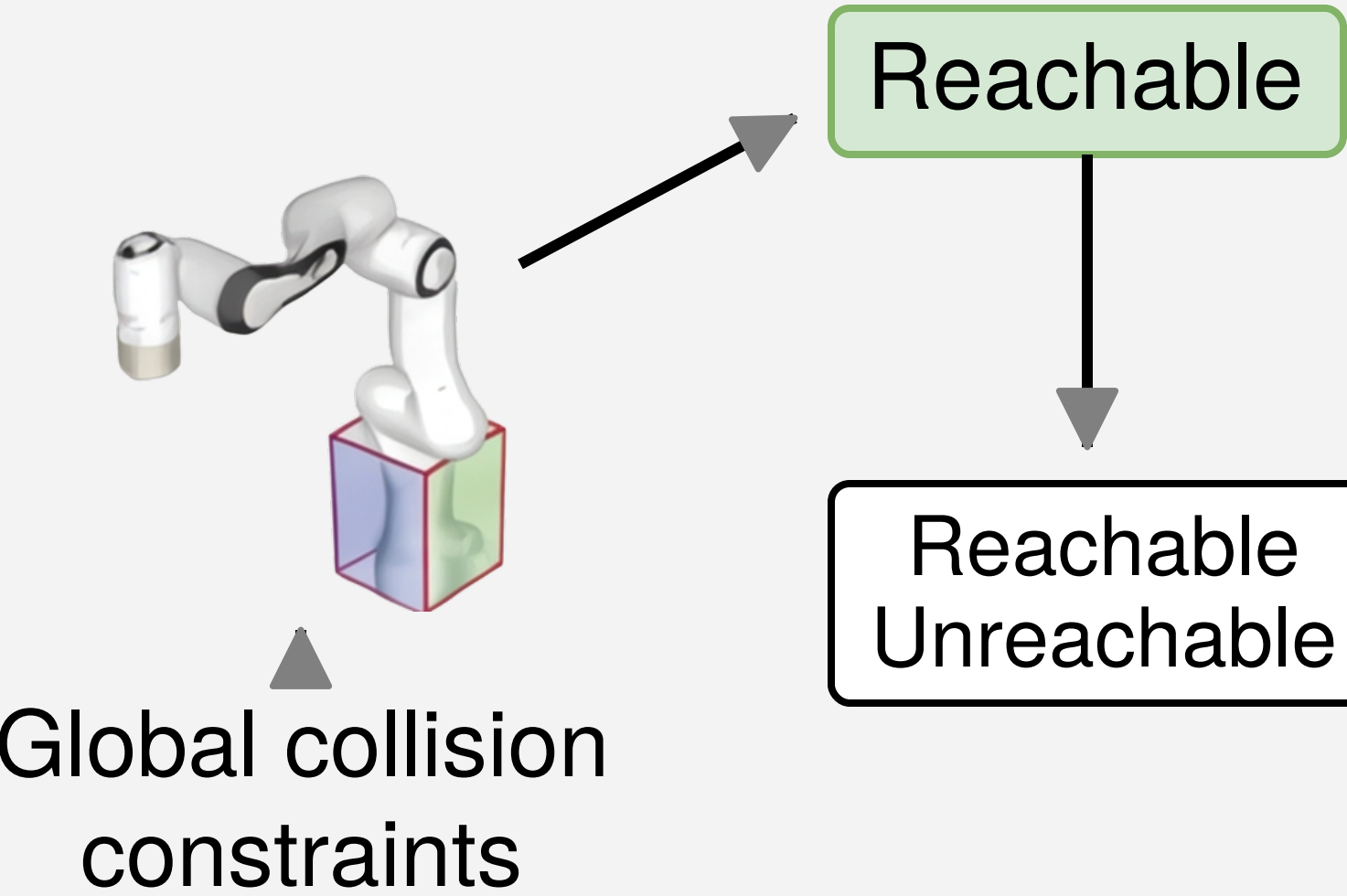
Input & Pre-check (Sec. 3.1)



Vision Module
(6D Poses)



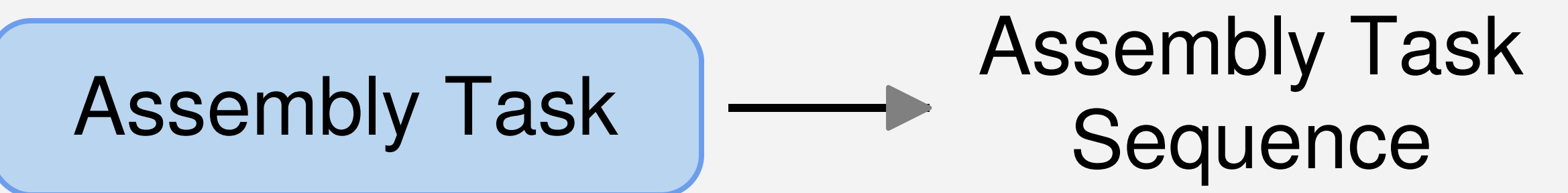
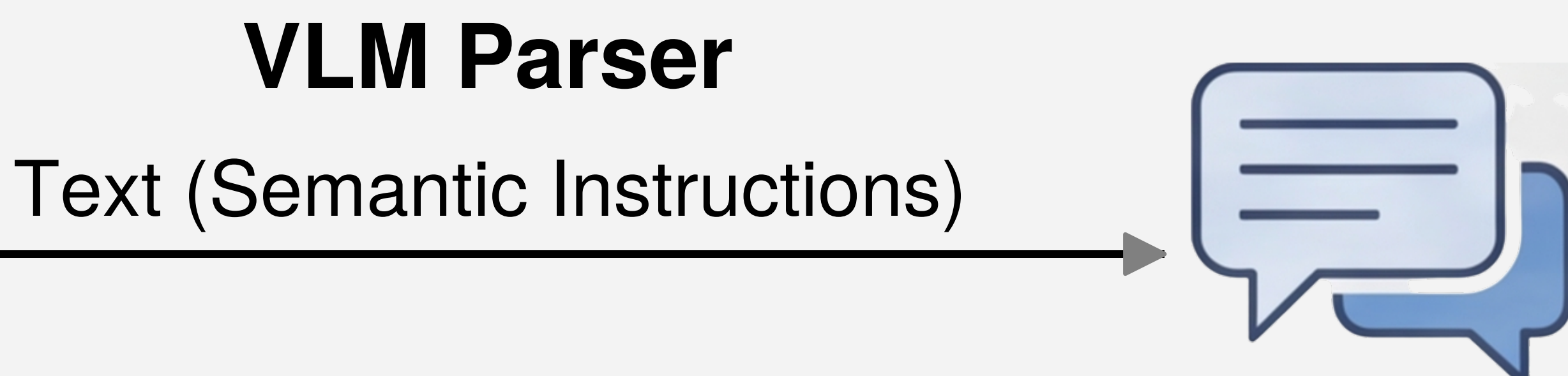
Spatial Sorting
(Euclidean Dist)



Reachability
Pre-check

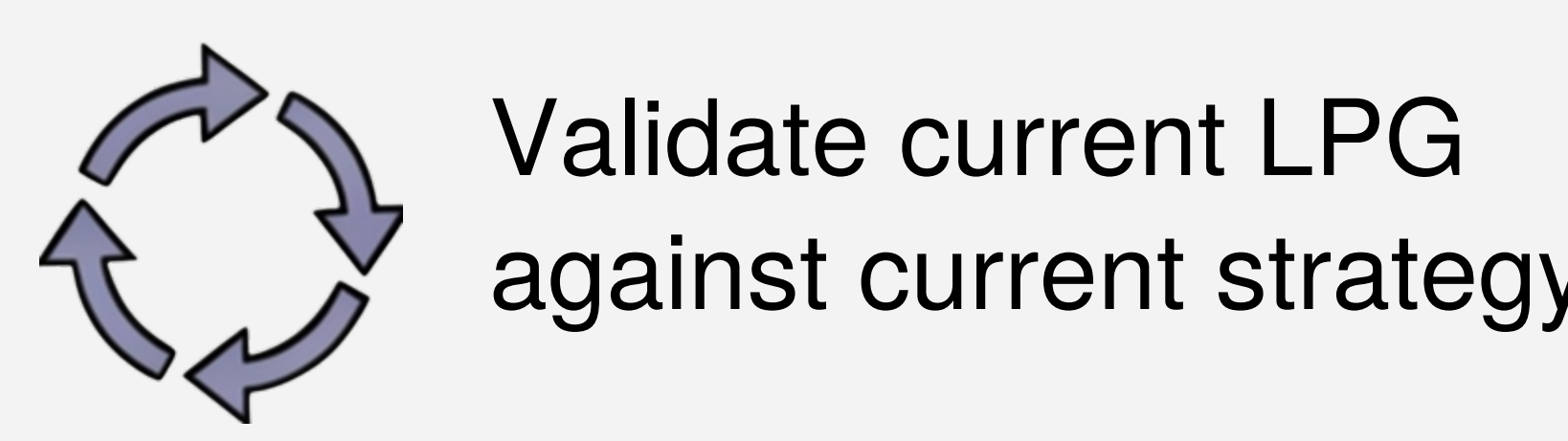
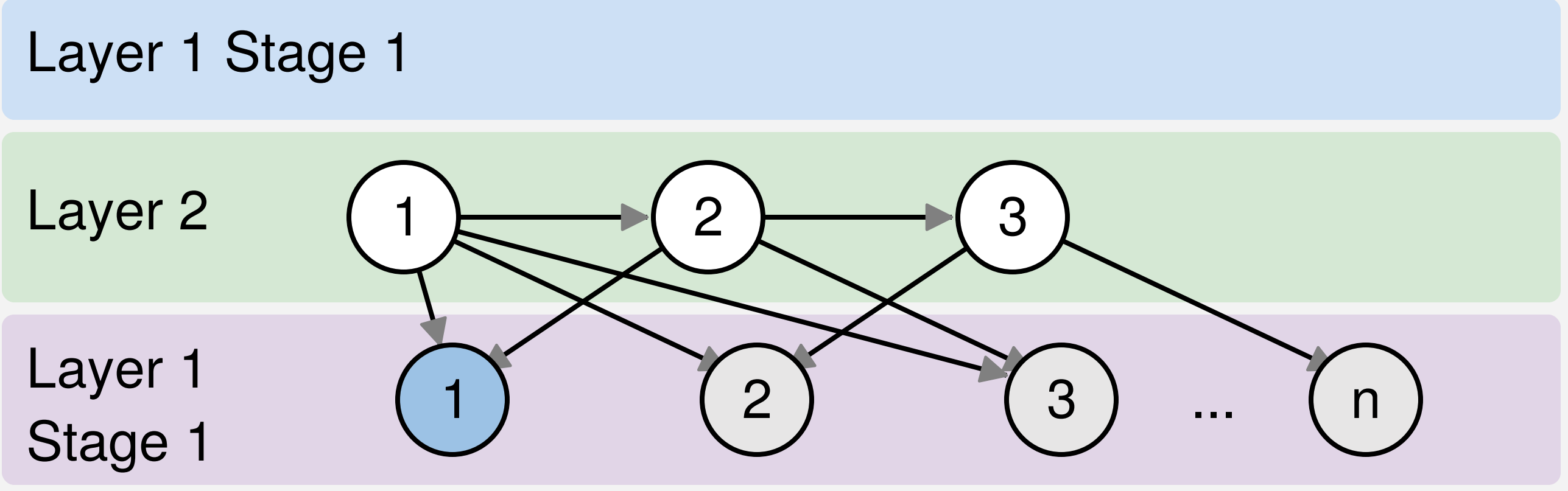
Framework (Sec. 3)

Layered Precedence (Sec. 3.2)



VLM Parser (Semantic Instructions)

Layered Precedence Graph (LPG)

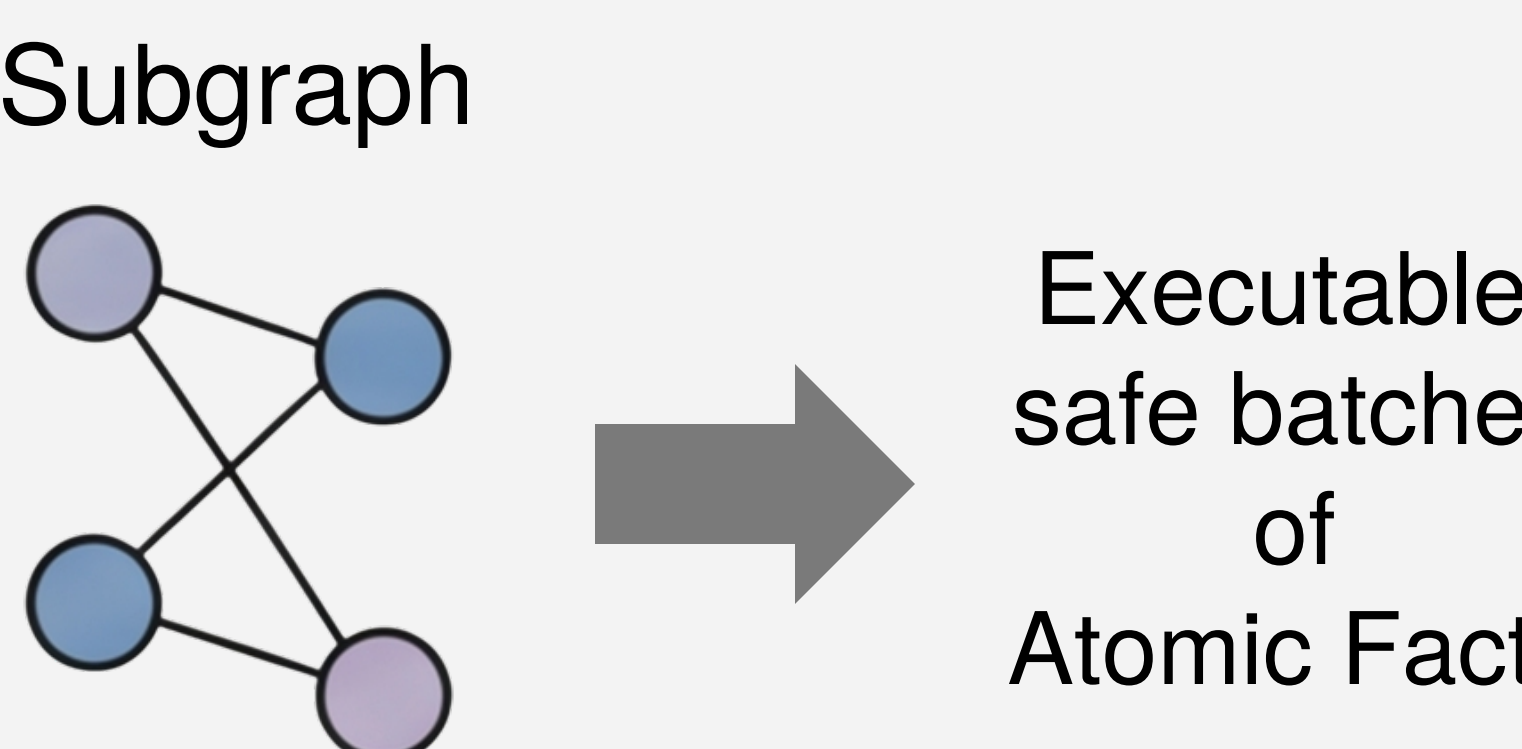
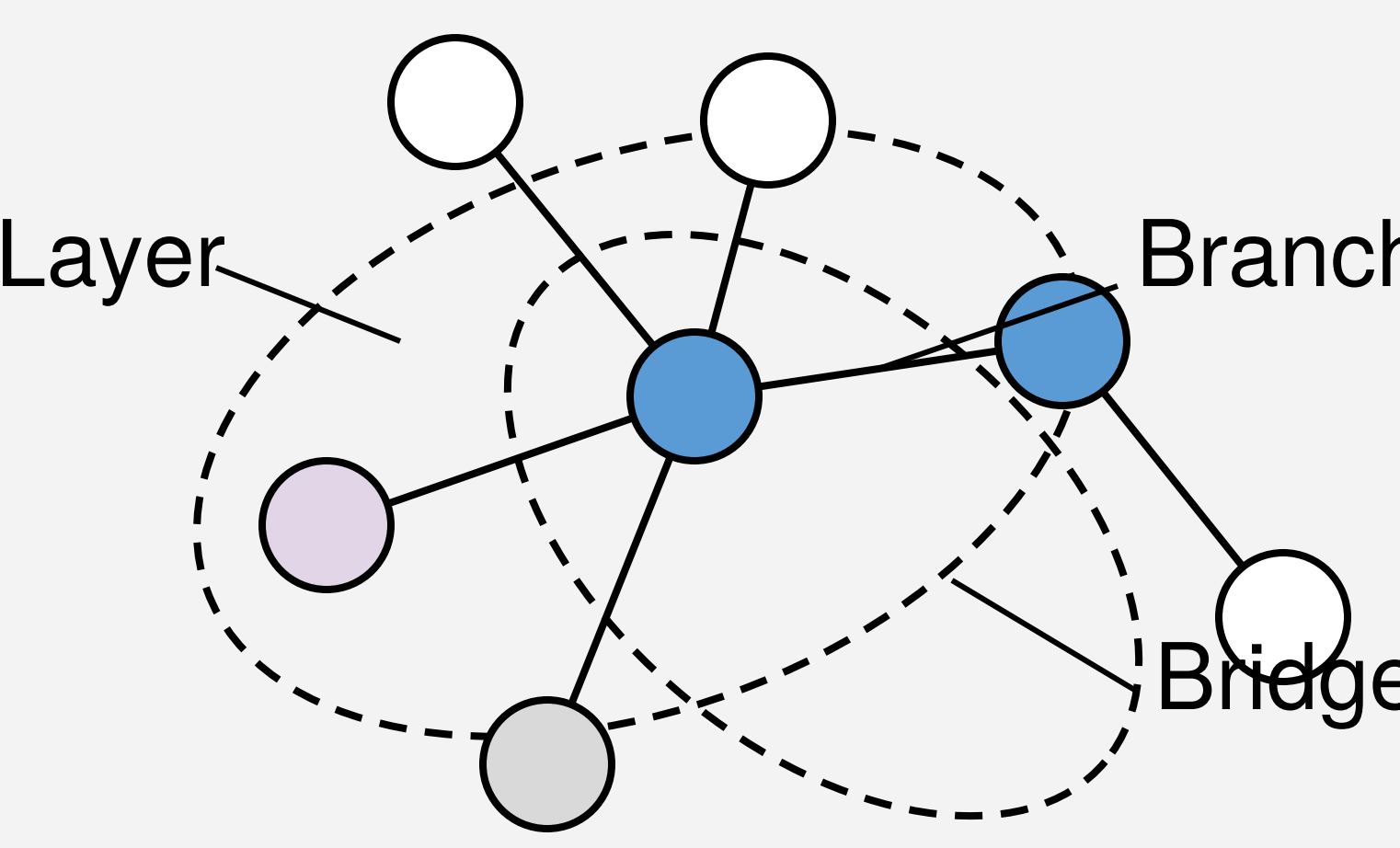


Python Compliance Monitor

Dynamic Pruning & Batching (Sec. 3.3)

Semantic Resource Allocator
Matches available physical instances to generic requests

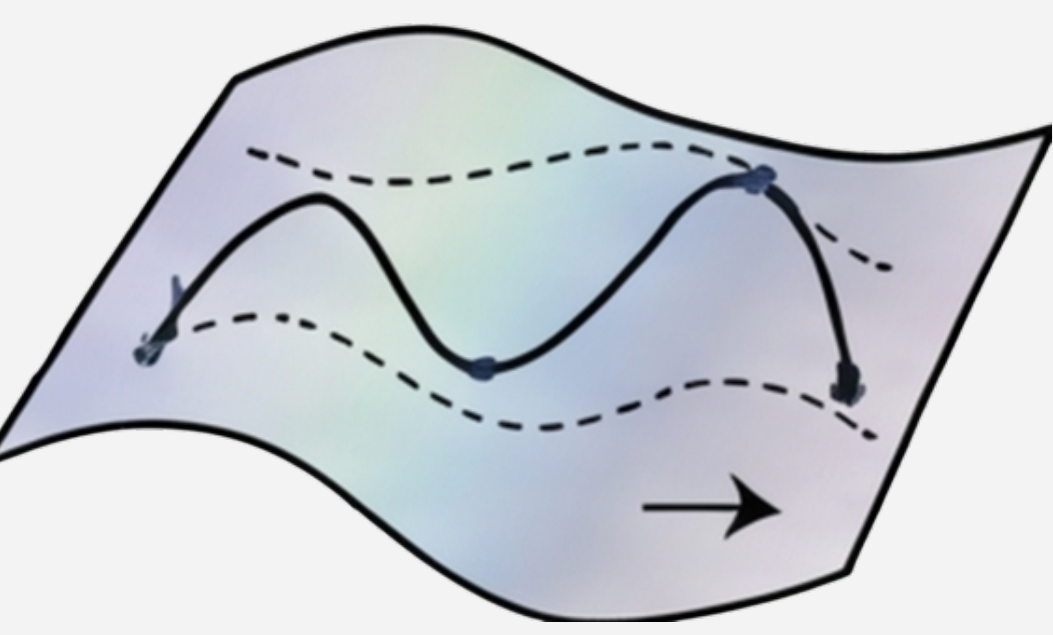
BATC Algorithm



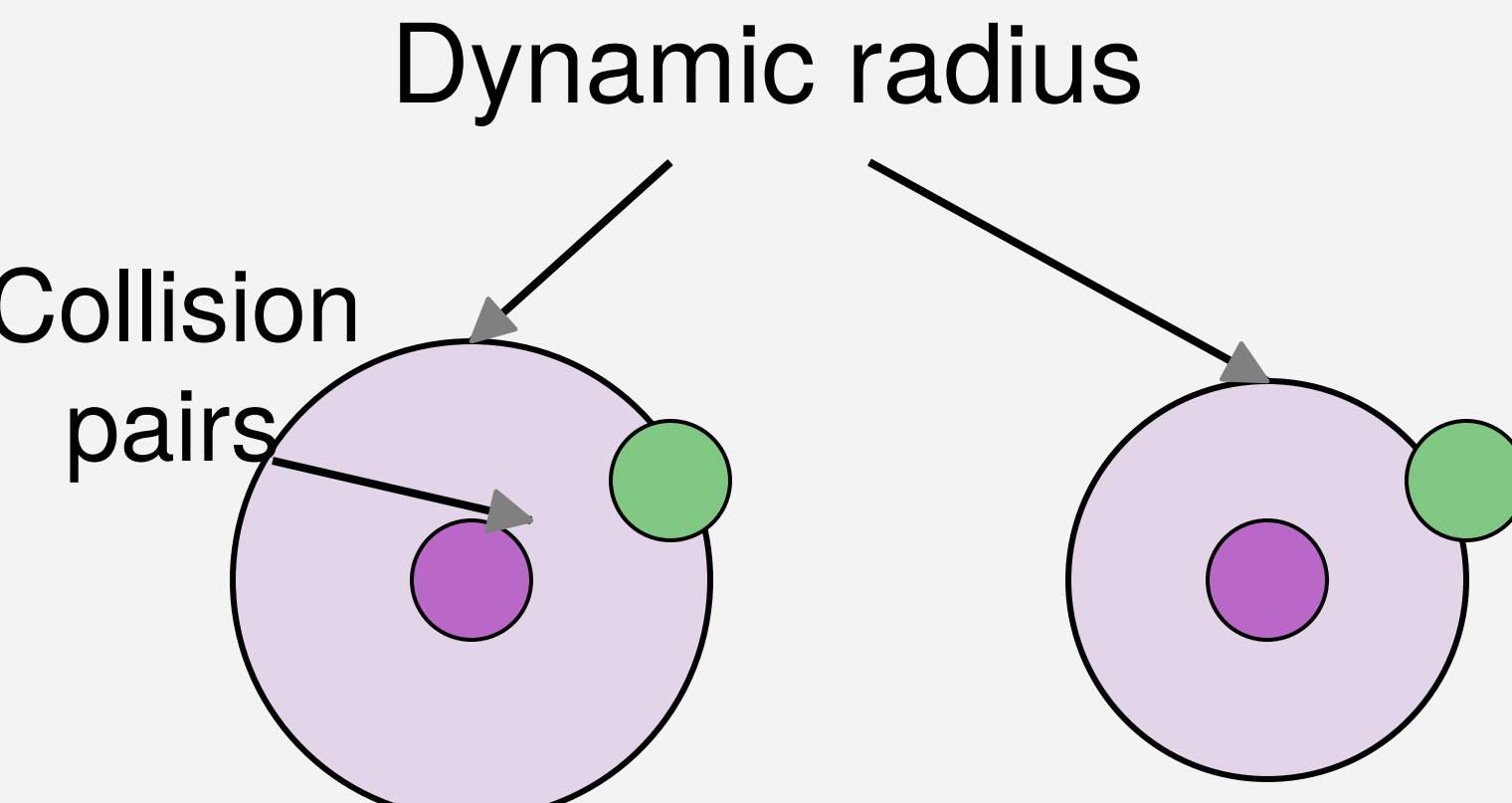
Logic Batch Compiler

Geometric Execution & Robustness (Sec. 3.4 & 3.5)

Trajectory Optimization
(k-Markov)

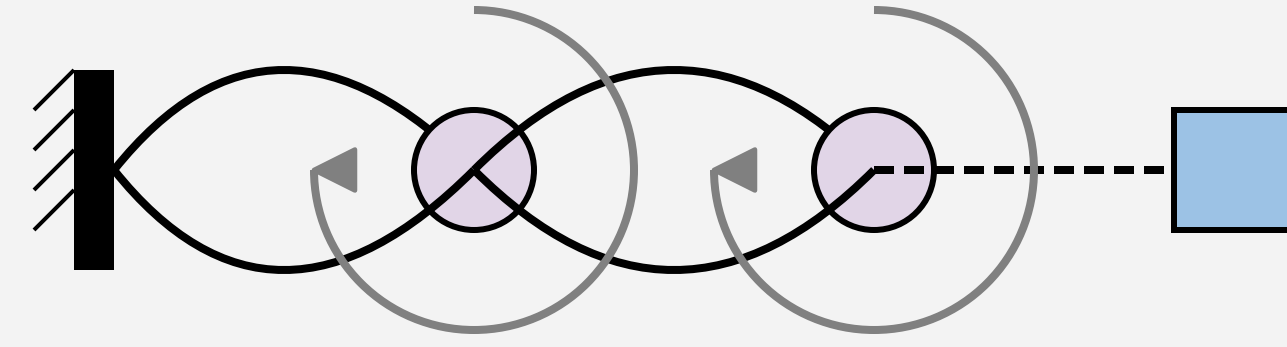


Trajectory Optimization
(k-Markov)



Active Constraint Strategy

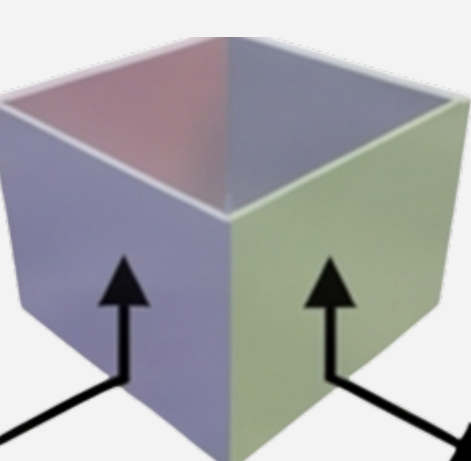
Breaking closed kinematic loop



Multi-support
Virtual Anchor

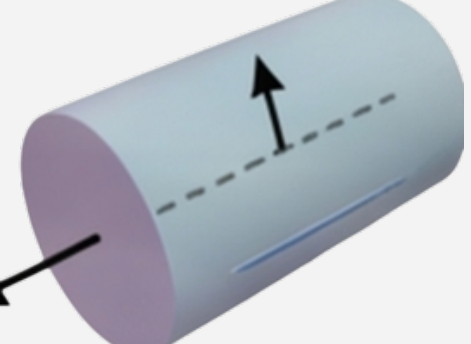
Manipulation Primitives

Box-type



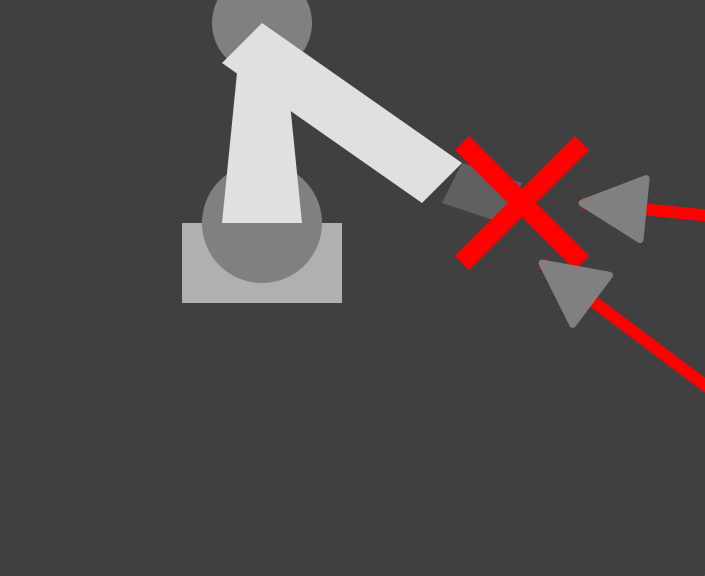
- Free Z-axis funnel
- XY-centered grasp, DoFs

Cylindrical

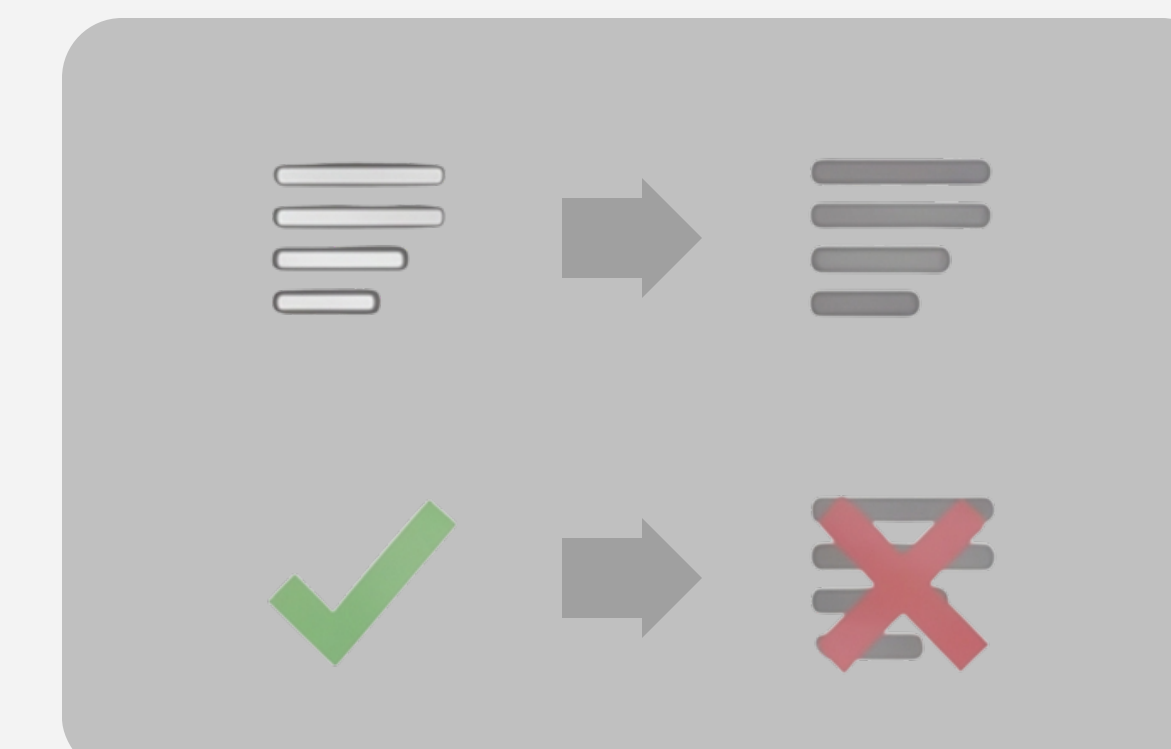


- Orthogonal approach axis
- Slide along central axis

Visual-Semantic Validator



Top-down visual capturon



Execution Traces
& Safety Halting